

4(a)	(two) waves travelling (at same speed) in opposite directions overlap	B1
	waves (are same type and) have same frequency/wavelength	B1
4(b)(i)	5	A1
4(b)(ii)	$T = 1/40 (= 2.5 \times 10^{-2})$	C1
	time taken = $2.5 \times 10^{-2} / 2$ $= 1.3 \times 10^{-2} \text{ s } (1.25 \times 10^{-2} \text{ s})$	A1
4(b)(iii)	180°	A1
4(b)(iv)	$v = f\lambda$	C1
	$\lambda = 2.0/2.5 (= 0.80 \text{ m})$ $v = 0.80 \times 40$ $= 32 \text{ m s}^{-1}$	A1